

OrBit Protocol: A Bitcoin-Collateralized Synthetic Currency Layer on Botanix EVM.

Bridging Bitcoin Capital to Local Fiat Liquidity without Selling Bitcoin.

orbitlabs.ng@gmail.com

Abstract

OrBit Protocol introduces a Bitcoin-collateralized synthetic currency system on Botanix EVM, providing Bitcoin holders with a mechanism to access local fiat denominated liquidity without selling their BTC holdings. This protocol enables users to lock Bitcoin as collateral and mint synthetic currency units that reference local fiat values, allowing fiat-denominated spending while preserving long-term Bitcoin exposure.

Initiated with a synthetic Nigerian naira unit (sNGN), OrBit Protocol is designed as a modular, multi-currency framework capable of supporting additional local currency units over time.

Drawing inspiration from Bitcoin's decentralized monetary design, Palladium Labs' credit-powered framework, and the technical innovations of Botanix EVM, OrBit Protocol positions itself as a pioneering infrastructure for the next frontier in Bitcoin finance.

1. Introduction

Bitcoin, as outlined in Nakamoto's seminal whitepaper, established a decentralized peer-to-peer monetary system designed to operate independently of traditional financial institutions. Despite global adoption, Bitcoin holders in emerging markets face unique liquidity challenges: while BTC is held as a long-term store of value, local fiat-denominated obligations require selling BTC or navigating complex, multi-step processes involving stablecoins like USDT and USDC, which introduces bridge friction, latency, and custody risk.

OrBit Protocol leverages the programmability of the Botanix EVM to create Bitcoin-collateralized synthetic tokens that reference the value of local fiat currencies. These synthetic tokens acts as a unit of account, allowing users to denominate value, manage local transactions, and integrate with existing DeFi and exchange ecosystems without selling Bitcoin.

2. The Need for Bitcoin Finance in Emerging Markets

2.1 Bitcoin as a Store of Value

For Bitcoin holders in emerging markets such as Nigeria, Bitcoin represents protection against currency volatility and macroeconomic instability. However, converting BTC to local fiat for real-world expenditures often involves:

1. Multiple bridging steps through stablecoins.
2. Exposure to counterparty risk.
3. Time delays and transaction fees.

These frictions discourage the use of Bitcoin-backed financial strategies in everyday life.

2.2 Liquidity Constraints

Bitcoin adoption is high but Bitcoin is not widely accepted as a medium of exchange by merchants and institutions thus, local fiat liquidity is essential for daily transactions such as paying rent, school fees, and medical expenses. Existing BTCFi systems are largely USD-centric and fail to provide seamless access to local currency-denominated liquidity without requiring selling Bitcoin off. This creates a structural barrier to broader Bitcoin adoption.

3. OrBit Protocol as a Bridging Innovation

OrBit Protocol addresses this liquidity gap through a Bitcoin-collateralized credit model that enables the minting of synthetic local currency units. Key features include:

1. Non-custodial over-collateralization: Users lock Bitcoin in a smart contract to mint synthetic units of account.
2. Unit of account functionality: Synthetic units allow users to price goods and services locally.
3. Direct integration with exchanges: Simplifies access to fiat liquidity without multiple bridging steps.
4. Credit-based design: Ensures Bitcoin remains the economic backing, preserving exposure while enabling temporary liquidity.

This mechanism bridges the gap between Bitcoin's role as a store of value and the need for fiat-denominated spending, allowing users to transact locally without liquidating BTC. The protocol is initiated with a synthetic NGN unit (sNGN) but architected to support additional currency units as the system evolves.

4. Architecture

OrBit Protocol builds on the technical foundations of Botanix EVM, leveraging:

1. Smart contract-based CDPs (Collateralized Debt Positions) for BTC-backed minting.
2. Synthetic currency tokens representing local units of account.
3. Automated liquidation and risk management mechanisms.
4. On-chain composability with DeFi infrastructure and exchange integrations.

Each synthetic currency unit operates with independent risk parameters, oracle feeds, and debt ceilings to ensure system resilience and prevent cross-currency contagion. Inspired by Palladium Labs' approach to credit-powered Bitcoin layers, OrBit maintains a decentralized, algorithmically governed peg while keeping all collateral fully on-chain.

5. Multi-Currency Design and Expansion

While OrBit Protocol is launched with a synthetic NGN unit, it is explicitly designed as a multi-currency platform. Additional synthetic units (e.g., sKES, sGHS, sZAR, sEUR) may be introduced over time, subject to governance approval and liquidity readiness. Each unit:

1. Is collateralized by Bitcoin.
2. References a specific local currency.
3. Maintains isolated risk and liquidation parameters.

This modular design allows OrBit to expand across jurisdictions while remaining Bitcoin-native and technically consistent.

6. Enabling Local Payments with Bitcoin Collateral

While USDT and USDC are commonly used in cross-border and local settlements, they introduce dependency on centralized issuers and require bridging. OrBit enables:

1. Bitcoin-backed local transactions: Synthetic currency units can be exchanged or integrated with local CEXs to facilitate fiat payments.
2. Preservation of Bitcoin exposure: Users maintain BTC as the underlying asset while meeting local fiat needs.
3. Rapid liquidity conversion: Eliminates multi-step USDT/USDC pathways.

This approach enhances Bitcoin adoption by positioning BTC not just as a speculative asset but as a directly usable economic resource in local financial systems.

7. Use Cases

1. Daily local spending: Pay for utilities, school fees, or rent using synthetic currency units backed by BTC.
2. DeFi composability: Use synthetic currency units within smart contracts for lending, trading, or local accounting.
3. Liquidity access without BTC liquidation: Meet up short term fiat needs without reducing long-term BTC holdings.

8. Conclusion

OrBit Protocol advances the frontier of Bitcoin finance by enabling Bitcoin-collateralized access to local fiat liquidity. Initiated with a synthetic NGN unit and designed for multi-currency expansion, OrBit bridges the gap between Bitcoin's role as a store of value and the realities of fiat-denominated economic activity.

By reducing reliance on USD stablecoins and centralized intermediaries, OrBit demonstrates how Bitcoin can directly underpin local financial systems and support practical adoption across emerging economies.

References

- [1] Nakamoto, S. Bitcoin: a *peer-to-peer electronic cash System*, 2008.
- [2] Palladium Labs. *Credit-powered Bitcoin layer: documentation*. 2025.
- [3] Botanix Labs. *Botanix: an EVM-equivalent layer built on Bitcoin*. Whitepaper. 2025.
- [4] MakerDAO. *Collateralized debt positions and synthetic assets*. 2023.
- [5] CoinDesk; Blockworks. *Coverage on Bitcoin-native stablecoins*. 2025.